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Market Edge Algorithm

Overview

Places a limit order to add liquidity at the edge of the order book, aiming for price improvement. If the order is not filled within a predefined time (e.g., 10 seconds), it is canceled and replaced with a market order to ensure immediate execution. The algorithm dynamically adjusts to market conditions to achieve the best price while ensuring the order is filled.

Detailed Explanation

The Market Edge Algorithm combines the benefits of both limit and market orders. It attempts to fill orders as a market maker first and switches to market orders if required for timely execution.

1. Order Placement Strategy:

- **Passive Orders:**

- Places a limit order at the edge of the order book to add liquidity

- **Price Calculation:**

For buy orders: $P_{\text{passive}} = P_{\text{bestask}}$

For sell orders: $P_{\text{passive}} = P_{\text{bestbid}}$

- **Aggressive Orders:**

- If the passive order is not fully filled within the allowed time (e.g., 10 seconds), it is canceled
- Places a market order to fill the remaining quantity immediately

2. Monitoring and Adjustments:

- The algorithm continuously tracks the filled quantity:

$$Q_{filled} = \sum_{fills} Q_{fill}$$

- If the order is not completely filled, the remaining quantity is calculated as:

$$Q_{remaining} = Q_{total} - Q_{filled}$$

3. Precision Adjustments:

- To meet exchange requirements, the algorithm ensures that both price and quantity are rounded to the instrument's specified precision
- For example, prices and quantities are rounded based on exchange-defined decimal places



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14 filled, 25 Not filled

- Sends a final status update and logs execution details

Supported Directives

Market-Edge supports several directives that modify its behavior to suit different trading scenarios:

Aggression Directives

Control how assertively the algorithm places orders:

- **aggression:** Object - Controls order placement strategy
 - `level` : String - Placement strategy level
 - `"aggressive"` : Places limit orders beyond best bid/ask for fastest execution
 - `"passive"` : Places limit orders at best bid/ask (default behavior)
 - `"neutral"` : Places limit orders at mid price

Example:

```
    "aggression": {
      "level": "passive"
    }
  },
  "json"
```

Time-Based Directives

Control the algorithm's timing behavior:

- **time:** Object - Controls timing parameters
 - `timeout_seconds` : Integer - Time to wait before switching to market order (default: 10)
 - `max_wait_time` : Integer - Maximum time the algorithm will run before completion

Example:

```
    "time": {
      "timeout_seconds": 15
    }
  },
  "json"
```

Trigger Directives

Define conditions for order execution:

- **trigger:** Object - Contains trigger conditions
 - **threshold_percent:** Object - Execute based on price deviation thresholds
 - `favorable_percent` : Decimal - Percent deviation that triggers execution when market moves favorably

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- `adverse_movement_percent` : Decimal - Percent move that triggers protective action
- `response` : String - Action to take (`"accelerate"` , `"abort"` , `"market"`)
- conditional_execution**: Object - Start execution only when conditions are met
 - `condition_type` : String - Type of condition (`"price"` , `"time"` , `"spread_ratio"`)
 - `condition_value` : Various - Value that triggers execution
 - `comparison` : String - Comparison operator (`"less_than"` , `"greater_than"` , `"equal_to"`)

Example:

```
json
{
  "trigger": {
    "threshold_percent": {
      "favorable_percent": 0.2,
      "adverse_percent": 0.5
    },
    "conditional_execution": {
      "condition_type": "price",
      "condition_value": 42500.5,
      "comparison": "less_than"
    }
  }
}
```

Execution Control Directives

Fine-tune execution behavior specific to market-edge:

- execution_control**: Object - Advanced execution parameters
 - `hedge_on_partial_fill` : Boolean - Whether to immediately hedge partial fills for paired instruments
 - `fill_acceleration` : Object - Controls adaptive execution based on fill rate
 - `enabled` : Boolean - Whether to enable fill acceleration
 - `threshold_percent` : Decimal - Fill percentage that triggers acceleration

Example:

```
json
{
  "execution_control": {
    "hedge_on_partial_fill": true,
    "fill_acceleration": {
      "enabled": true,
      "threshold_percent": 40
    }
  }
}
```

Liquidity Awareness Directives

Adjust execution based on available liquidity:

- liquidity**: Object - Controls liquidity-responsive behavior



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Example:

```
"liquidity": {  
  "responsive": true,  
  "depth_levels": 4,  
  "min_size": 0.01  
}
```

Implementation Example

Here's an example of how to use directives with the Market-Edge algorithm:

```
{  
  "algorithm": "market_edge",  
  "instrument": "BTC-USDT",  
  "side": "buy",  
  "quantity": 1.5,  
  "directives": {  
    "aggression": {  
      "level": "passive" // "User-defined Aggression Levels" - placing at best  
    },  
    "time": {  
      "timeout_seconds": 15, // "Time-based execution" - passive phase duration  
      "max_wait_time": 600 // "Total execution time" - 10 minutes total duration  
    },  
    "trigger": {  
      "threshold_percent": {  
        "favorable_percent": 0.2, // "Favorable Ratio Threshold" - execute when  
        "adverse_percent": 0.5 // "Adverse Movement (%)" - protection threshold  
      },  
      "adverse_movement": {  
        "adverse_movement_percent": 0.5, // "Adverse Movement Trigger" - 0.5%  
        "response": "accelerate" // "Aggressive execution on adverse moves"  
      },  
      "conditional_execution": {  
        "condition_type": "spread_ratio", // "Conditional Limit Order Execution"  
        "condition_value": 42.6, // "Favorable ratio trigger threshold"  
        "comparison": "less_than", // "For BUY spread, execute when <= threshold"  
        "instrument_pair": "BTC-USDT-PERP/ETH-USDT-PERP" // "Spread pair trading"  
      }  
    },  
    "execution_control": {  
      "hedge_on_partial_fill": true, // "When $35 of B fills, immediately execute"  
      "fill_acceleration": {  
        "enabled": true, // "Volume-based Execution" - accelerate based on fill  
        "threshold_percent": 40 // "Accelerate if 40% filled" - adaptation threshold  
      }  
    }  
  },  
  "liquidity": {  
    "responsive": true, // "Adapt to available liquidity on order book"  
    "depth_levels": 4, // "Consider 4 levels of depth on asset A"  
    "min_size": 0.01 // "Minimize partial fills" - set minimum execution size  
  }  
}
```



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Use Cases

Basic Execution

Scenario: A trader wants to buy **1,000 shares** of ABC stock and optimize the execution price.

Approach:

- The trader uses the Market Edge Algorithm with a timeout of **10 seconds**
- The algorithm places a passive limit order at the edge of the order book, adding liquidity
- If the order is not fully filled within **10 seconds**, the passive order is canceled
- The remaining quantity is executed using a market order, ensuring the order is completed

Outcome:

- The trader achieves optimal pricing for the shares filled passively
- The remaining shares are filled immediately at the market price
- Combines price improvement opportunity with guaranteed execution

Spread Trading

Market-Edge is especially valuable for the entry and exit legs of spread trading strategies:

- 1. Entry Execution:** When opening a spread position
 - Use threshold_percent directives to execute only when the spread moves favorably
 - Apply aggression directives to optimize execution price
- 2. Exit Execution:** For take profit or stop loss
 - Define adverse movement directives to protect against rapid market movements
 - Use conditional execution to trigger only when specific profit or loss thresholds are met
 - Set max_wait_time to ensure positions are closed within a defined timeframe

Liquidity-Responsive Execution

For trading pairs with asymmetric liquidity (where one asset is more liquid than another):

- 1. Initial Approach:**
 - Places a passive limit order on the less liquid asset
 - Uses the liquidity.responsive directive to evaluate order book depth of the more liquid asset
 - Sets liquidity.depth_levels to determine how many price levels to consider when evaluating available liquidity



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- After the timeout period expires, executes any remaining quantity using a market order
- Uses liquidity.min_size to avoid placing orders too small to be economically viable

Accelerated Volume-Based Execution

For executing large orders in volatile markets:

1. **Gradual Approach:**
 - Begin with passive limit orders to minimize market impact
 - Set fill_acceleration.enabled to true to dynamically adjust execution pace
2. **Adaptive Acceleration:**
 - When fill_acceleration.threshold_percent (e.g., 40%) of the order fills successfully at passive prices
 - The algorithm automatically accelerates execution for remaining quantity
 - Helps achieve optimal balance between waiting for favorable prices and timely completion

Conditional Market Timing

For entering positions only under specific market conditions:

1. **Price-Based Conditions:**
 - Set conditional_execution with condition_type "price" to enter only when an instrument reaches a target price
 - Prevents executing in unfavorable price environments
2. **Spread Ratio Conditions:**
 - For pair trading, use condition_type "spread_ratio" with comparison operators
 - Example: Execute BTC/ETH spread trade only when ratio falls below 13.5
 - Ensures entry at statistically significant deviation points
3. **Time-Based Conditions:**
 - Schedule execution around known market events or liquidity windows
 - Particularly useful for markets with distinct trading sessions or announcement periods

Summary

The Market Edge Algorithm is ideal for traders seeking:

- **Price Optimization:** By initially placing limit orders at favorable prices
- **Guaranteed Execution:** By switching to market orders if necessary
- **Dynamic Adaptability:** Continuous monitoring of market conditions ensures efficient execution
- **Customizable Behavior:** Extensive directive support allows for precise control over execution dynamics



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